Multistatic SAR Sequential Imaging Method Using Subaperture Spatial Spectrum Fusion Based on High-Precision Trajectory Reconstruction

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***Abstract*—This article focuses on simultaneously achieving a short synthetic aperture time and high imaging resolution to improve the observation capability of microwave-band synthetic aperture radar (SAR). To this end, multistatic SAR (MSAR) for sequential imaging is investigated. Through the distribution and collaboration of multiple radars, MSAR synthesizes a long azimuth aperture much faster than conventional SAR. However, insufficient positioning accuracy of the MSAR platforms introduces significant errors in imaging and leads to the failure of high-resolution multistatic image formation. A key challenge lies in how to coherently fuse echoes from MSAR under multiplatform trajectory errors. To address this issue, we propose an MSAR sequential imaging method using subaperture spatial spectrum fusion based on high-precision trajectory reconstruction, so as to ensure imaging accuracy. In this method, multiplatform multiorder trajectory errors in MSAR sequential imaging are modeled, and a high-precision multiplatform trajectory reconstruction method is developed based on the quantitative relationship between subaperture image offsets and trajectory error parameters. Subsequently, an MSAR sequential imaging framework is established to compensate for multiplatform trajectory errors and coherently fuse subaperture images from all radars, thereby improving image resolution. For validation, a three-uncrewed aerial vehicle (UAV) MSAR system is established, and a sequential imaging experiment is conducted. The effectiveness of the proposed method is validated using real measurements, demonstrating an enhancement in imaging resolution.**

***Index Terms*—Coherent data fusion, multistatic synthetic aperture radar (MSAR), resolution improvement, spectrum fusion, trajectory reconstruction.**

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I. Introduction

SYNTHETIC aperture radar (SAR) offers all-weather, all-day remote sensing capabilities over long distances, finding extensive applications in both military and civilian sectors [1], [2], [3], [4], [5], [6]. Driven by technologies such as high-precision time and space synchronization and multiplatform formation control, multistatic SAR (MSAR) has gradually attracted the attention of researchers. MSAR system [7], [8], [9], [10], [11] offers enhanced survivability and superior concealment. Moreover, due to its highly flexible configuration, MSAR has more spatial degrees of freedom (DOFs) and offers superior imaging performance. A typical example is that it can tradeoff between high resolution and short synthetic aperture of microwave-band SAR [12], [13], [14].

In traditional microwave-band SAR, higher resolution demands a longer synthetic aperture length. Given a fixed platform velocity, this implies a longer synthetic aperture time. Improving image resolution without lengthening synthetic aperture time remains a key challenge. Overlapping apertures technology can address this issue. It is able to shorten the synthetic aperture time of microwave-band SAR [15], [16], and different frames of images have considerable coherence. While its frame rate improvement relies on azimuth data overlapping, high redundancy of image information from different frames exists.

MSAR sequential imaging offers a solution to this challenge [12], [13], [14]. In MSAR sequential imaging, the distributed short apertures along azimuth are synthesized individually, and equivalent large apertures are synthesized collectively by all platforms. To be more specific, MSAR decomposes the traditional SAR's aperture synthesis process spatially, and the entire synthesis aperture is divided into parts, each of which is synthesized by the corresponding radar in MSAR. Therefore, MSAR sequential imaging guarantees the entire aperture length and reduces the aperture synthesis time in the meantime. In this way, MSAR sequential imaging possesses shorter aperture synthesis time, which enhances SAR imaging capability and is appreciated in civil and military applications.

In recent years, several scholars have conducted research on sequential imaging for MSAR. Ding et al. [13] proposed a high-frame-rate imaging method based on

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orthogonal waveforms for multiple self-transmitting self-receiving MSAR systems. This method adapts signal parameters to variations in imaging geometry and achieves MSAR image fusion in the spatial spectral domain. The study further analyzes the advantages of MSAR high-frame-rate imaging in moving target indication and imaging. Wang et al. [17] introduced a fast MSAR image formation method that simultaneously performs motion error phase compensation. Its effectiveness is validated using multipass MSAR measurements. Meanwhile, Jiang et al. [18] investigated the influence of sparse aperture on along-track MSAR imaging and employed compressive sensing to recover more azimuth data, thereby improving image resolution. Furthermore, Jiang et al. [19] presented a method for MSAR moving target parameter estimation and image formation. It validates the superior applicability of MSAR sequential imaging in moving target parameter estimation and explores MSAR sequential imaging's advantages in dynamic observation of interested areas. In summary, while existing research has covered MSAR image formation, phase error estimation, sparse aperture image formation, and moving target parameter estimation, studies on MSAR trajectory error modeling, trajectory error compensation, and image formation under trajectory error conditions remain insufficient.

The limited accuracy of current navigation equipment makes the available trajectory measurements far from sufficient for monostatic SAR imaging, let alone for the more demanding MSAR sequential imaging. Trajectory errors introduce range migration errors and phase errors to MSAR echoes. Unlike traditional monostatic SAR, MSAR's multiple distributed platforms determine that its range migration error and phase error are discontinuous and segmented, tremendously deteriorating the imaging results and bringing significant difficulties for error compensation. Reconstructing the trajectory is an ideal way to ensure imaging accuracy and avoid the influence of trajectory errors. Several monostatic SAR and bistatic SAR accurate trajectory reconstruction and high-precision imaging methods have been proposed. Li et al. [20] extracted strong scatterers' Doppler to invert bistatic SAR trajectory, which cannot reconstruct linear trajectory error. However, linear trajectory error components of each platform contribute to the MSAR's overall high-order trajectory error, which cannot be neglected. In addition, solving trajectory error point-by-point is time-consuming. Zhong et al. [21], [22] solved for the actual trajectory based on the relationship of subaperture images' phase difference and trajectory error parameters, but they required isolated strong points distributed in azimuth. A series of isolated strong point targets distributed in azimuth is a stringent condition for imaging scenes. In addition, selecting isolated strong point targets is not easily automated. While in [23], isolated strong point targets are not necessary to solve trajectory deviation. It calculates the subaperture image offset using image registration and reconstructs the trajectory based on the relationship between image offsets and the trajectory error parameters, which ensures imaging accuracy. However, the low-order error model in [21] and [23] considers intercept and slope only, which cannot completely describe the MSAR's distributed platforms' trajectory error. To realize MSAR multiplatform trajectory reconstruction, not only the errors of each platform's trajectory need to be reconstructed but also global multiplatform trajectory reconstruction is essential. In conclusion, due to model mismatch and computational burden, existing trajectory reconstruction methods cannot be directly applied to MSAR. There is a lack of precise modeling and reconstruction methods for MSAR multisource distributed trajectory error.

To tackle the above difficulties, an MSAR sequential imaging method using subaperture spatial spectrum fusion based on high-precision trajectory reconstruction is proposed in this article. First, according to MSAR sequential imaging geometry and trajectory error analysis, the trajectory error model for each platform dominated by linear error, in other words, low-order errors, is developed. Based on the trajectory error model, a quantitative relationship between multiplatform low-order trajectory error parameters and MSAR's each platform's coarse-resolution image offset is deduced. Then, the multiple-trajectory error reconstruction method is proposed. In the method, each platform's high-order and low-order trajectory errors are reconstructed one by one. The high-order trajectory errors are estimated by maximum-sharpness autofocus (MSA), ensuring each platform's coarse-resolution focused image. The low-order trajectory errors are reconstructed based on the quantitative relationship between trajectory error parameters and coarse-resolution image offset. The reconstructed error-free trajectory lays the foundation for high-precision MSAR sequential imaging. At last, the MSAR sequential imaging method is proposed, in which subaperture coarse-resolution images of all radars in MSAR are formed in the spatial spectrum domain and fused according to their actual spatial spectrum distribution. Thus, compensation of multiplatform trajectory errors and coherent fusion of subaperture images from all radars are accomplished. For validation of the proposed method, an MSAR sequential experiment is conducted, and the imaging results are given, demonstrating the resolution improvement ability of the proposed method.

The remainder of this article is organized as follows. Section II constructs the MSAR sequential imaging echo model and analyzes the influence of trajectory errors. The proposed sequential imaging method is presented in Section III. Section IV shows the numerical simulations and experimental results of the proposed algorithms. Finally, conclusions are drawn in Section V.

II. MSAR Sequential Imaging Model

This section comprehensively studies the concepts and geometry of MSAR sequential imaging, analyzes its performance statistically, and establishes the MSAR sequential imaging echo model and trajectory error model, and then investigates the influence of trajectory error on MSAR sequential imaging.

A. MSAR Sequential Imaging Configuration

Fig. 1 shows MSAR sequential imaging geometry. The MSAR sequential imaging system contains a total of N radars. Also, the N radars are distributed in azimuth. Their trajectories

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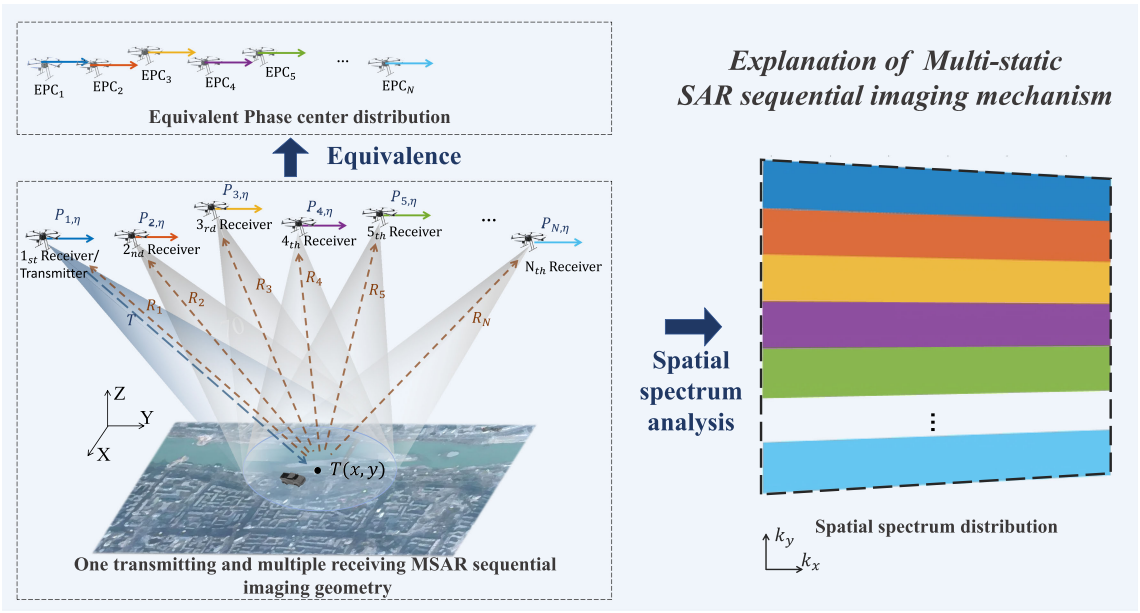


Fig. 1. MSAR sequential imaging mechanism.

are represented by segments in different colors. Meanwhile, the first radar is equipped with transmit and receive antennas, while the other radars are only equipped with receive antennas.

In MSAR sequential imaging, all platforms simultaneously move along azimuth at a constant speed, in spotlight mode. The transmitter transmits a wideband linear frequency modulation (LFM) signal to the target area with a fixed pulse repetition interval (PRI), and all receivers receive echoes from the scene. Suppose $\boldsymbol{T}$ is a target with coordinate (x_T, y_T). At azimuth time η, the nth receiver's position is denoted as $\boldsymbol{P}_{n,\eta}$, and the range history of target $\boldsymbol{T}$ consists of range history from first receiver to target $\boldsymbol{T}$, and range history from the nth receiver to target $\boldsymbol{T}$, which is denoted as

$$R_{n,\eta}^{bi} = R_{1,\eta} + R_{n,\eta} \tag{1}$$

where $R_{n,\eta} = \|\boldsymbol{P}_{n,\eta}\boldsymbol{T}\|$ represents the range history from the nth receiver to target $\boldsymbol{T}$.

The essence that MSAR sequential imaging can acquire high resolution and short synthetic aperture time simultaneously is that multiple radars in MSAR synthesizing a large number of short apertures distributing in azimuth, which is equivalent to monostatic SAR synthesizing an equal length long aperture. As illustrated in Fig. 1, though all platforms in MSAR have distributed trajectories, their equivalent phase center distribution is continuous in azimuth, where EPC_n denotes the equivalent phase center distribution of the transmitter and the nth receiver [18], [24]. Furthermore, the results of the MSAR spatial spectral distribution analysis are given, and its spatial spectrum is also 2-D continuous. The relationship between the point spread function and spatial spectrum can be expressed by the 2-D Fourier transform. Spatial spectrum distribution decides the shape of the point spread function. MSAR sequential imaging geometry has a spatial spectrum distribution consistent with that of long-aperture monostatic SAR, thus revealing the same resolution ability.

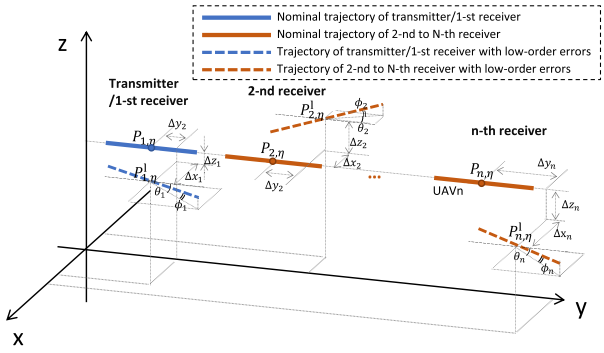


Fig. 2. MSAR low-order trajectory error model.

B. Trajectory Error Model

In actual measurements, errors in trajectory measurement are inevitable due to the insufficient accuracy (typically limited to 1–10 cm) of the differential global positioning system (DGPS) or inertial navigation system (INS). This section analyzes the MSAR trajectory error's specificities and deduces the error model accordingly.

The actual MSAR trajectory is modeled as the sum of the trajectory with only low-order trajectory errors, and high-order trajectory errors. Thus, it can be modeled as

$$\boldsymbol{P}'_{n,\eta} = \boldsymbol{P}^{l}_{n,\eta} + \boldsymbol{P}''_{n,\eta} \tag{2}$$

where $\boldsymbol{P}'_{n,\eta} = (x'_{n,\eta}, y'_{n,\eta}, z'_{n,\eta})$ denotes the UAVn's actual location, $\boldsymbol{P}^{l}_{n,\eta} = (x^{l}_{n,\eta}, y^{l}_{n,\eta}, z^{l}_{n,\eta})$ is the UAVn's location under low-order trajectory error, and $\boldsymbol{P}''_{n,\eta} = (x''_{n,\eta}, y''_{n,\eta}, z''_{n,\eta})$ is the UAVn's high-order trajectory errors. MSAR's trajectory under low-order trajectory error $\boldsymbol{P}^{l}_{n,\eta}$ is shown in Fig. 2.

It can be seen that the MSAR trajectory with low-order errors is modeled as several line segments. The nominal trajectories are represented by solid line segments, and the trajectory with low-order errors is depicted by dotted line

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segments. Low-order errors are expressed in terms of the translational and rotational components in 3-D space. The 3-D translation of trajectory center is represented by $\Delta \boldsymbol{x}$, $\Delta \boldsymbol{y}$ and $\Delta \boldsymbol{z}$, and the trajectory rotation caused by uncrewed aerial vehicle (UAV) velocity direction offset is described by roll angle $\boldsymbol{\phi}$ and yaw angle $\boldsymbol{\theta}$

$$\begin{aligned}\Delta \boldsymbol{x} &= [\Delta x_1, \Delta x_2, \ldots, \Delta x_n] \\ \Delta \boldsymbol{y} &= \left[\Delta y_1, \Delta y_2, \ldots, \Delta y_n\right] \\ \Delta \boldsymbol{z} &= [\Delta z_1, \Delta z_2, \ldots, \Delta z_n] \\ \boldsymbol{\theta} &= [\theta_1, \theta_2, \ldots, \theta_n] \\ \boldsymbol{\phi} &= \left[\phi_1, \phi_2, \ldots, \phi_n\right]. \end{aligned} \tag{3}$$

According to the geometrical relationship, UAVn's position under low-order error $\boldsymbol{P}_{n,\eta}^{l} = (x_{n,\eta}^{l}, y_{n,\eta}^{l}, z_{n,\eta}^{l})$ and its nominal position $\boldsymbol{P}_{n,\eta} = (x_{n,\eta}, y_{n,\eta}, z_{n,\eta})$ can be expressed as

$$\begin{bmatrix} x_{n,\eta}^{l} \\ y_{n,\eta}^{l} \\ z_{n,\eta}^{l} \end{bmatrix} = \boldsymbol{M}_{\phi_n} \boldsymbol{M}_{\theta_n} \begin{bmatrix} x_{n,\eta} \\ y_{n,\eta} \\ z_{n,\eta} \end{bmatrix} + \begin{bmatrix} \Delta x_n \\ \Delta y_n \\ \Delta z_n \end{bmatrix} \tag{4}$$

where

$$\boldsymbol{M}_{\phi_n} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi_n & \sin\phi_n \\ 0 & -\sin\phi_n & \cos\phi_n \end{bmatrix} \tag{5}$$

and

$$\boldsymbol{M}_{\theta_n} = \begin{bmatrix} \cos\theta_n & \sin\theta_n & 0 \\ -\sin\theta_n & \cos\theta_n & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{6}$$

Target's actual range history is $R'_{n,\eta} = \|\boldsymbol{P}'_{1,\eta}\boldsymbol{T}\| + \|\boldsymbol{P}'_{n,\eta}\boldsymbol{T}\|$, and the range history error is derived from the difference between the actual distance range and the nominal range history error

$$\Delta R_{n,\eta}^{bi} = R_{n,\eta}^{bi'} - R_{n,\eta}^{bi}. \tag{7}$$

According to the MSAR geometry given in Fig. 1, the demodulated echo of the nth receiver is expressed as

$$\begin{aligned}\mathbf{S}_n(\tau,\eta) = \sigma_{\boldsymbol{T}} \operatorname{rect}\left(\frac{\eta}{T_a}\right) \operatorname{rect}\left(\frac{\tau - R_{n,\eta}^{bi}/c}{T_r}\right) \\ \times \exp\left(-j\frac{2\pi f_c R_{n,\eta}^{bi}}{c} + j\pi K\left(\tau - \frac{R_{n,\eta}^{bi}}{c}\right)^2\right)\end{aligned} \tag{8}$$

where $\sigma_{\boldsymbol{T}}$ denotes the radar cross section (RCS) of scatterer $\boldsymbol{T}$, rect$(\cdot)$ represents the window function, τ and η are the fast time and the azimuth time, respectively, T_a is the synthetic aperture time, T_r indicates the pulsewidth, f_c is the carrier frequency, K denotes the frequency modulation rate, and c represents the light speed.

The nth receiver's echo after range pulse compression is

$$\begin{aligned}\mathbf{S}_n^{rc}(\tau,\eta) = \sigma_{\boldsymbol{T}} \operatorname{rect}\left(\frac{\eta}{T_a}\right) \operatorname{sinc}\left(B_r\left(\tau - \frac{R_{n,\eta}^{bi}}{c}\right)\right) \\ \times \exp\left(-j\frac{2\pi f_c R_{n,\eta}^{bi}}{c}\right)\end{aligned} \tag{9}$$

where B_r denotes the bandwidth and MSAR's echo is composed of echoes of all receivers.

Under the influence of trajectory error, echo of the nth receiver after pulse compression is written as

$$\begin{aligned}\boldsymbol{S}_n^{rc}(\tau,\eta) = \sigma_{\boldsymbol{T}} \operatorname{rect}\left(\frac{\eta}{T_a}\right) \operatorname{sinc}\left(B\left(\tau - \frac{R_{n,\eta}^{bi} + \Delta R_{n,\eta}^{bi}}{c}\right)\right) \\ \times \exp\left(-j\frac{2\pi f_c\left(R_{n,\eta}^{bi} + \Delta R_{n,\eta}^{bi}\right)}{c}\right).\end{aligned} \tag{10}$$

Range history error $\Delta R_{n,\eta}^{bi}$ caused by multiplatform trajectory error brings phase and envelope errors into MSAR echo, deteriorating MSAR sequential image quality.

C. Trajectory Error Analysis

Spatially discontinuous trajectory errors are modeled in Section II-B. It is tough to analyze and compensate for the MSAR distributed trajectory error. Therefore, this section separately analyzes the effects of high-order trajectory errors and low-order trajectory errors on MSAR imaging.

There is no essential difference between the effects of high-order trajectory errors on conventional monostatic SAR and MSAR, which have been analyzed in detail in the literature [14], [20], [21], [23]. High-order trajectory errors introduce envelope and phase errors that vary with azimuth time and deteriorate the image quality.

Following, how the low-order trajectory error affects image results is analyzed in two steps. First, the effect of trajectory error on each receiver's subaperture image is analyzed. Then, the influence of trajectory errors on MSAR full-resolution imaging is analyzed, which lays a foundation for subsequent high-precision imaging.

1) Low-Order Trajectory Error's Influence on Each Receiver's Image: The impact of low-order trajectory errors on MSAR receiver imaging is investigated by studying the effects of low-order errors from both the transmitter and the receiver on the receiver's imaging quality.

Both the transmitter's and receiver's trajectory errors lead to the target's range history error, and the range history error degrades the image results. To further investigate the impact of low-order trajectory errors in imaging, we conducted 1000 Monte Carlo simulations. These simulations statistically analyzed the bistatic range histories of the transmitter and receiver under varying low-order trajectory errors to calculate the magnitude of each order's component. The transmitter and receiver are lined up along the track, traveling at the same speed, 200 m apart and on a 53-m trajectory. The range between the transmitter and the target is 5000 m. The translation and rotation of the trajectories are set under the state-of-the-art 10-cm-level positioning accuracy.

The maximum and minimum of each order component of the range history error in 1000 experiments are given in Table I. It shows that linear components dominate range history errors, and high-order components can be ignored. Therefore, multiplatform trajectory error leads to the position offset of every receiver's subaperture image, and the offset amount is related to the corresponding receiver's trajectory error.

To demonstrate the relationship between image offset and corresponding trajectory error, Fig. 3 plots position variations

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TABLE I
RANGE HISTORY ERROR ANALYSIS OF EACH PLATFORM IN MSAR

Components of range history error	Minimum	Maximum
linear component	2.04×10^{-5} m	0.55 m
2nd order component	5.06×10^{-17} m	5.56×10^{-12} m
3rd order component	2.18×10^{-16} m	3.46×10^{-12} m

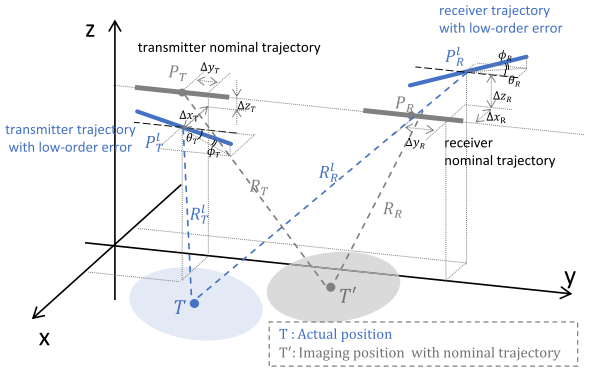


Fig. 3. Position variations of the target in the presence of low-order trajectory error.

of the target in the presence of low-order trajectory error. The gray solid line is the ideal trajectory of the transmitter and receiver, and the blue solid line is their actual trajectory with low-order error. The error of transmitter trajectory and receiver trajectory are represented by $(\Delta x_{Tx}, \Delta y_{Tx}, \Delta z_{Tx}, \theta_{Tx}, \phi_{Tx})$ and $(\Delta x_{Rx}, \Delta y_{Rx}, \theta_{Rx}, \phi_{Rx})$. $\boldsymbol{T}$ represents the target's actual position, and $\boldsymbol{T}'$ represents the image position in the presence of low-order trajectory errors.

The range history equation can be derived according to the range history center of nominal position $\boldsymbol{T}$ and offset position $\boldsymbol{T}'$ due to trajectory error

$$R_{Tx} + R_{Rx} = R'_{Tx} + R'_{Rx} \tag{11}$$

where $R_{Tx} = \|\boldsymbol{P}^l_{Tx} - \boldsymbol{T}\|$ and $R_{Rx} = \|\boldsymbol{P}^l_{Rx} - \boldsymbol{T}\|$ denote the distance from $\boldsymbol{T}$ to the center of transmitter and receiver trajectory with low-order trajectory errors, and $R'_{Tx} = \|\boldsymbol{P}_{Tx} - \boldsymbol{T}'\|$ and $R'_{Rx} = \|\boldsymbol{P}_{Rx} - \boldsymbol{T}'\|$ represent the distance from $\boldsymbol{T}'$ to the center of nominal transmitter and receiver trajectory, respectively.

Also, the Doppler frequency centroid equation can be derived accordingly

$$\frac{\boldsymbol{v}^l_{Tx}\left(\boldsymbol{P}^l_{Tx} - \boldsymbol{T}\right)^T}{\lambda\|\boldsymbol{P}^l_{Tx} - \boldsymbol{T}\|} + \frac{\boldsymbol{v}^l_{Rx}\left(\boldsymbol{P}^l_{Rx} - \boldsymbol{T}\right)^T}{\lambda\|\boldsymbol{P}^l_{Rx} - \boldsymbol{T}\|} = \frac{\boldsymbol{v}_{Tx}\left(\boldsymbol{P}_{Tx} - \boldsymbol{T}'\right)^T}{\lambda\|\boldsymbol{P}_{Tx} - \boldsymbol{T}'\|} + \frac{\boldsymbol{v}_{Rx}\left(\boldsymbol{P}_{Rx} - \boldsymbol{T}'\right)^T}{\lambda\|\boldsymbol{P}_{Rx} - \boldsymbol{T}'\|} \tag{12}$$

where $\boldsymbol{v}^l_{Tx}$ and $\boldsymbol{v}^l_{Rx}$ describe the actual flight velocity vector of the transmitter and receiver, respectively, $\boldsymbol{v}_{Tx}$ and $\boldsymbol{v}_{Rx}$ indicate the nominal velocity vector of the transmitter and receiver, $\boldsymbol{P}_{Tx}$ and $\boldsymbol{P}_{Rx}$ refer to the center of nominal transmitter and receiver trajectory, $\boldsymbol{P}'_{Tx}$ and $\boldsymbol{P}'_{Rx}$ are the center of transmitter's and receiver's trajectory with low-order errors, λ is the wavelength, and T represents the transpose operation. Thus, the left side of (12) is the actual Doppler centroid of target $\boldsymbol{T}$, and the right side is the Doppler centroid of target $\boldsymbol{T}'$.

Equations (11) and (12) characterize the quantitative relationship between the image's offset position and the low-order error parameters of the platform's trajectory. Thus, low-order trajectory error affects the position of each receiver's image but does not affect the image's focus.

2) Low-Order Trajectory Error's Influence on MSAR Full-Resolution Image: Based on the above analysis, MSAR low-order trajectory errors result in an offset of each receiver's coarse-resolution image. Coherent fusing all receivers' coarse-resolution images in MSAR can form a full-resolution image. However, unlike traditional monostatic SAR, MSAR's range history error is discontinuous and segmented. Each segment has a different slope and intercept. Without an actual trajectory, the coherent fusion of all receivers' images will lead to target energy dispersion. Thus, the MSAR full-resolution image will defocus.

The effect of low-order trajectory error on MSAR sequential imaging is demonstrated by a 16-platform MSAR image simulation. Low-order trajectory error compatible with the state-of-the-art measurement accuracy of 10 cm is added. A point target is set at coordinate (0, 0, 0) m. Among the 16 platforms, the coarse-resolution imaging results of two platforms are shown in Fig. 4(a) and (b).

It can be observed that, under the influence of low-order trajectory errors, the images from the third and 15th receivers exhibit noticeable 2-D offsets. In contrast, when no low-order trajectory errors are present, the coarse images from these two platforms show no positional offset in Fig. 4(e) and (f). Furthermore, Fig. 4(c) and (g) presents the 3-dB contour lines of imaging results from all platforms with and without trajectory errors, respectively. It is evident that, in the presence of errors, the 3-dB contours from different platforms exhibit distinct 2-D shifts, and the magnitude of these shifts correlates with the error components. Conversely, in the error-free case, the 3-dB contours from all platforms align without positional offset; only a rotation angle, determined by each platform's squint angle, is observed. Fig. 4(d) and (h) displays the final MSAR reconstructed images with and without trajectory errors, respectively. These results clearly demonstrate that the low-order trajectory errors from individual platforms collectively introduce an equivalent high-order error into the MSAR system. This degrades coherence between platforms and causes severe defocusing in the final image. Therefore, to achieve high-quality full-resolution MSAR imagery, it is essential to compensate for multiplatform low-order trajectory errors with high precision during the imaging processing stage.

III. MSAR SEQUENTIAL IMAGING METHOD

In this section, to achieve high-precision MSAR sequential imaging, an MSAR sequential imaging method using subaperture spatial spectrum fusion based on high-precision trajectory reconstruction is proposed.

In the proposed method, first, all receivers' echoes are imaged with backprojection to obtain a series of coarse-resolution images. The coarse-resolution images exhibit

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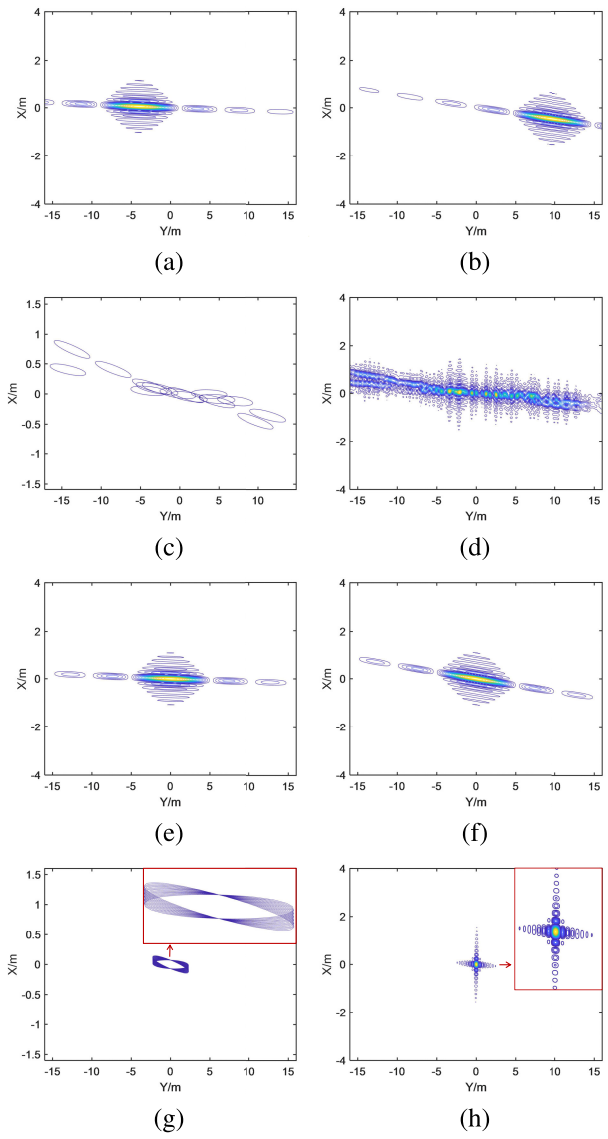


Fig. 4. 16-platform MSAR imaging result with and without low-order trajectory error. (a) Image of third receiver with trajectory error. (b) Image of 15th receiver with trajectory error. (c) 3-dB contours of 16 receivers with trajectory error. (d) Fusion image of all receivers with trajectory error. (e) Image of third receiver without trajectory error. (f) Image of 15th receiver without trajectory error. (g) 3-dB contours of 16 receivers without trajectory error. (h) Fusion image of all receivers without trajectory error.

defocusing and offsets caused by trajectory errors. Then, the multiplatform trajectory reconstruction is performed. In this step, autofocus is first adopted to estimate each platform's high-order trajectory error. Second, the image registration is completed to obtain the offset of each receiver's image. Third, the actual trajectory error parameters of the multiplatforms in MSAR are solved according to the relationship between trajectory error parameters and image offset. Finally, using the reconstructed trajectory, not only are the receivers' echoes reimaged, but also their spatial spectral distributions are computed. Subsequently, coherent fusion is performed in the spatial spectrum domain based on these actual distributions. The flowchart of the proposed method is given in Fig. 5.

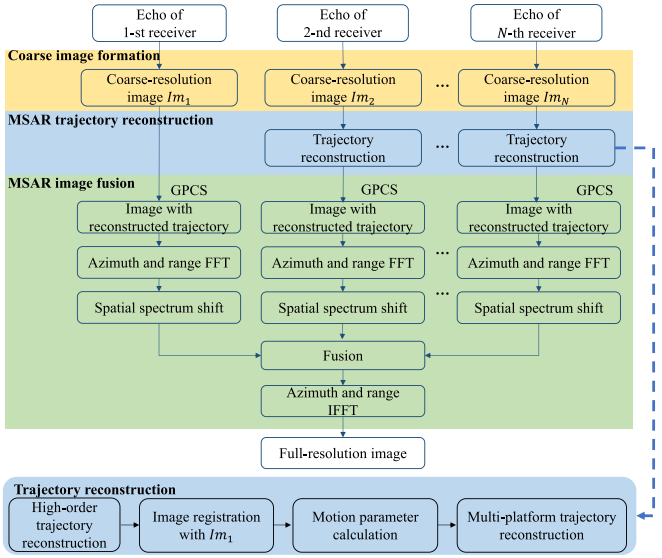


Fig. 5. Proposed MSAR sequential imaging method using subaperture spatial spectrum fusion based on high-precision trajectory reconstruction.

A. Coarse Image Formation

Coarse-resolution imaging of every receiver's echo is carried out. The nth receiver's coarse-resolution image is generated as

$$\boldsymbol{I}_n(x,y) = \int_{-\frac{T_a}{2}}^{\frac{T_a}{2}} \boldsymbol{S}_n^{rc}(\tau,\eta) \cdot \exp\left(j\frac{2\pi f_c R_{n,\eta}^{bi}(x,y)}{c}\right) d\eta \quad (13)$$

where $R_{n,\eta}^{bi}(x,y) = R_{1,\eta}(x,y) + R_{n,\eta}(x,y)$ denotes the range history of target (x,y).

Due to the mismatch between the nominal range history $R_{n,\eta}^{bi}$ and the actual range history $R_{n,\eta}^{bi} + \Delta R_{n,\eta}^{bi}$ in (10), the imaging results from each receiver are degraded by both high-order and low-order trajectory errors of the platform. As a result, not only is the imaging resolution unable to meet the preset requirements, but image offsets also remain uncorrected.

B. MSAR Trajectory Reconstruction

For each receiver's coarse-resolution imaging, high-order trajectory error affects MSAR image focus, while low-order trajectory error causes image offset. Thus, high-precision trajectory reconstruction can be divided into two main steps. First step is to recover each receiver's high-order trajectory error, in the meantime, ensure image focus quality. Second, each receiver's image offset information is used to solve the trajectory low-order error parameters. Ultimately, the MSAR trajectory error is wholly reconstructed.

1) High-Order Trajectory Estimation: According to the autofocus method [23], [25], with the aim of the optimal degree of image focus, the high-order trajectory error estimation can be obtained for optimally estimating the high-order phase error of each platform echo.

Then, suppose the phase error induced by high-order trajectory error $x''_{n,\eta}, y''_{n,\eta}, z''_{n,\eta}$ is denoted as $\phi''_{n,\eta}$, the nth receiver's

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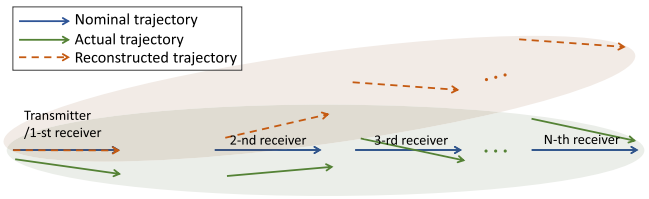


Fig. 6. Schematic of low-order trajectory error reconstruction.

compensated imaging result is expressed as

$$\boldsymbol{I}'_n(x,y) = \int_{-\frac{T_a}{2}}^{\frac{T_a}{2}} \boldsymbol{S}_n^{im} e^{-j\phi''_{n,\eta}} d\eta \tag{14}$$

where $\boldsymbol{S}_n^{im} = \boldsymbol{S}_n^{rc}(\tau,\eta)\exp(j2\pi f_c R_{n,\eta}^{bi}(r,\theta)/c)$, according to (13).

It is assumed that after the trajectory error is correctly estimated and compensated, the image achieves good focus, and the sharpness of the image is optimal, so the mathematical expression of the optimization problem is obtained

$$\hat{\boldsymbol{\Phi}} = \arg\max_{\boldsymbol{\Phi}} L(\boldsymbol{\Phi}) \tag{15}$$

where $L(\boldsymbol{\Phi})$ is the image sharpness, defined as $L(\boldsymbol{\Phi}) = \sum_{p=1}^{N_p} f_p^2, p = 1,2,3,\ldots,N_p$, where p is image $\boldsymbol{I}'_n$'s pixel index, and $f_p = z_p z_p^*$ is the intensity of pth pixel z_p. Ash [25] deduced a concise solution of (15) using the coordinate descent method; thus, the detailed solution steps are not repeated here. Then, the estimated phase $\boldsymbol{\Phi}$ can be mapped to an equivalent line-of-sight (LOS) range error using the relation $\boldsymbol{\Phi} = 2\pi\Delta R''/\lambda$, and then projected onto the LOS plane to reconstruct the trajectory error geometrically.

Thus, the MSAR high-order trajectory error is estimated by solving each receiver's image maximum sharpness problem.

2) Low-Order Trajectory Estimation: Unless there are persistent scatterers with exactly known positions, the actual position of the targets usually cannot be obtained. Then, in (11) and (12), only the imaging position with error $\boldsymbol{T}'$ is known. At the same time, the actual position $\boldsymbol{T}$ of the target and the trajectory errors of the receiving and transmitting stations are unknown, so it is difficult to estimate the errors. It is considered that the linear trajectory error will lead to target offset but will not cause target defocus. Therefore, this article adopts the idea of inverting the trajectory error of each receiver relative to the transmitter by taking the transmitter as the reference and reconstructing the error of each receiver relative to the transmitter one by one to reconstruct the high-order error of the overall equivalent long aperture and ensure the focusing effect of the target.

As illustrated in Fig. 6, the nominal trajectories are plotted by blue segments, taking the first receiver (i.e., the transmitter)'s trajectory as a reference and reconstructing other trajectories. It is capable of reconstructing the relative position of all receivers' trajectories, as plotted by orange segments. Therefore, there are still overall position shifts and rotations between the nominal trajectory and reconstructed trajectory, which is represented by the ellipses, with the orange ellipse indicating the range of the reconstructed trajectory and the green ellipse indicating the range of the actual trajectory. Such overall shift and rotation do not affect the image focusing and are omitted in this article.

The principle of solving the trajectory error of the nth receiver is illustrated below. Taking the position of the first receiver as the reference, that is, the trajectory of the first receiver is considered as the actual trajectory without low-order error, then $\Delta x_1 = 0$, $\Delta y_1 = 0$, $\Delta z_1 = 0$, $\phi_1 = 0$, and $\theta_1 = 0$. Thus, the target $\boldsymbol{T}$'s position in the image of the first receiver is taken as its actual position. Equations (11) and (12) are basis of trajectory reconstruction of the nth receiver. Rewriting them according to the first receiver and the nth receiver's configuration, replacing $\boldsymbol{T}$ and $\boldsymbol{T}'$ with $\boldsymbol{T}_1$ and $\boldsymbol{T}_n$ denotes

$$R_1^l + R_n^l = R_1 + R_n \tag{16}$$

where R_1^l and R_n^l denote the distance from actual position $\boldsymbol{T}_1$ to the center of the first receiver's trajectory and the nth receiver's trajectory under low-order error, respectively. Also, R_1 and R_n represent the distance from offset position $\boldsymbol{T}_n$ to the center of first and the nth receiver's nominal trajectory, respectively,

$$\frac{\boldsymbol{v}_1(\boldsymbol{P}_1-\boldsymbol{T}_1)^T}{\lambda\|\boldsymbol{P}_1-\boldsymbol{T}_1\|} + \frac{\boldsymbol{v}_n^l\left(\boldsymbol{P}_n^l-\boldsymbol{T}_1\right)^T}{\lambda\|\boldsymbol{P}_n^l-\boldsymbol{T}_1\|} = \frac{\boldsymbol{v}_1(\boldsymbol{P}_1-\boldsymbol{T}_n)^T}{\lambda\|\boldsymbol{P}_1-\boldsymbol{T}_n\|} + \frac{\boldsymbol{v}_n(\boldsymbol{P}_n-\boldsymbol{T}_n)^T}{\lambda\|\boldsymbol{P}_n-\boldsymbol{T}_n\|} \tag{17}$$

where $\boldsymbol{v}_n^l$ and $\boldsymbol{v}_n$ describe the nth receiver's actual and nominal flight velocity vector, respectively. Also, $\boldsymbol{P}_n^l$ and $\boldsymbol{P}_n$ refer to center of actual and nominal trajectory of the nth receiver.

Then, in (16) and (17), only the nth receiver position with low-order errors, denoted as $\boldsymbol{P}_n^l(x_n,y_n,z_n)$, and the rotation amount of the velocity vector θ_n and ϕ_n are unknown. For simplicity, let $R_1 + R_n - R_1^l = r_n$, $(\boldsymbol{v}_1(\boldsymbol{P}_1-\boldsymbol{T}_n)^T)/(\lambda\|\boldsymbol{P}_1-\boldsymbol{T}_n\|) + (\boldsymbol{v}_n(\boldsymbol{P}_n-\boldsymbol{T}_n)^T)/(\lambda\|\boldsymbol{P}_n-\boldsymbol{T}_n\|) - (\boldsymbol{v}_1(\boldsymbol{P}_1-\boldsymbol{T}_1)^T)/(\lambda\|\boldsymbol{P}_1-\boldsymbol{T}_1\|) = D_n$, (16) and (17) can be rewritten as

$$R_n^l = r_n \tag{18}$$

$$\frac{\boldsymbol{v}_n^l\left(\boldsymbol{P}_n^l-\boldsymbol{T}_1\right)^T}{\lambda\|\boldsymbol{P}_n^l-\boldsymbol{T}_1\|} = D_n \tag{19}$$

where r_n and D_n are the constants decided by nominal trajectory parameters and imaging position of all receivers' images. r_n characterizes the range between target position $\boldsymbol{T}$ and the nth receiver, and D_n characterizes the Doppler centroid due to the relative motion of target position $\boldsymbol{T}$ and the nth receiver. Then, expand R_n^l as the distance between $\boldsymbol{P}_n^l(x_n^l,y_n^l,z_n^l)$ and $\boldsymbol{T}(x_{\boldsymbol{T}_1},y_{\boldsymbol{T}_1},z_{\boldsymbol{T}_1})$, then

$$\sqrt{\left(x_n^l-x_{\boldsymbol{T}_1}\right)^2+\left(y_n^l-y_{\boldsymbol{T}_1}\right)^2+\left(z_n^l-z_{\boldsymbol{T}_1}\right)^2} = r_n. \tag{20}$$

Equation (20) has the interpretation of a sphere, indicating that the real position of the nth receiver's trajectory is on the sphere with $\boldsymbol{T}_1$ as the center and r_n as the radius.

The first step of retrieving the trajectory error is to obtain the real position of the target point and the image position after offset. In this article, the SAR-SIFT image registration algorithm [26] is used to estimate offsets between the image

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of the nth receiver and the first receiver, denoted as $\mathbf{IM}_n$ and $\mathbf{IM}_1$

$$\begin{bmatrix} x_{\boldsymbol{T}_n} \\ y_{\boldsymbol{T}_n} \\ 1 \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{\boldsymbol{T}_1} \\ y_{\boldsymbol{T}_1} \\ 1 \end{bmatrix} = \boldsymbol{F} \begin{bmatrix} x_{\boldsymbol{T}_1} \\ y_{\boldsymbol{T}_1} \\ 1 \end{bmatrix} \tag{21}$$

where $\boldsymbol{F}$ indicates the affine transformation relationship between the two images, and $\boldsymbol{T}_1(x_{\boldsymbol{T}_1}, y_{\boldsymbol{T}_1})$ and $\boldsymbol{T}_n(x_{\boldsymbol{T}_n}, y_{\boldsymbol{T}_n})$ denote the any target position in image of the first receiver and the nth receiver, respectively.

Following, since SAR imaging is performed on the ground plane, the elevation of the targets can be assumed to be 0, i.e., $z_{\boldsymbol{T}} = 0$. According to only three points $\boldsymbol{T}_1^1(x_{\boldsymbol{T}_1^1}, y_{\boldsymbol{T}_1^1})$, $\boldsymbol{T}_1^2(x_{\boldsymbol{T}_1^2}, y_{\boldsymbol{T}_1^2})$, and $\boldsymbol{T}_1^3(x_{\boldsymbol{T}_1^3}, y_{\boldsymbol{T}_1^3})$ in image of the first receiver and their corresponding position in image of the nth receiver, $\boldsymbol{T}_n^1(x_{\boldsymbol{T}_n^1}, y_{\boldsymbol{T}_n^1})$, $\boldsymbol{T}_n^2(x_{\boldsymbol{T}_n^2}, y_{\boldsymbol{T}_n^2})$, and $\boldsymbol{T}_n^3(x_{\boldsymbol{T}_n^3}, y_{\boldsymbol{T}_n^3})$, the distance equations are listed and simplified according to (20)

$$\begin{cases} \left(x_n^l - x_{\boldsymbol{T}_1^1}\right)^2 + \left(y_n^l - y_{\boldsymbol{T}_1^1}\right)^2 + \left(z_n^l\right)^2 = \left(r_n^1\right)^2 \\ \left(x_n^l - x_{\boldsymbol{T}_1^2}\right)^2 + \left(y_n^l - y_{\boldsymbol{T}_1^2}\right)^2 + \left(z_n^l\right)^2 = \left(r_n^2\right)^2 \\ \left(x_n^l - x_{\boldsymbol{T}_1^3}\right)^2 + \left(y_n^l - y_{\boldsymbol{T}_1^3}\right)^2 + \left(z_n^l\right)^2 = \left(r_n^3\right)^2. \end{cases} \tag{22}$$

The system of equations above represents the equations of three spheres, and its solution $\boldsymbol{P}_n$ corresponds to the intersection points of these spheres. By subtracting the sphere equations pairwise, the quadratic terms $(x_n^l)^2$,$(y_n^l)^2$,$(z_n^l)^2$ are eliminated, yielding two linear plane equations. Solving for the line of intersection of these two planes and substituting it back into any of the sphere equations in (22) allows for the determination of the intersection points $\boldsymbol{P}_n$ of the three spheres.

It should be noted that the three spheres usually have two points of intersection, and the false point $\boldsymbol{P}_n^{fake}$ exists. It is easy to avoid its interference. On the one hand, in fact, the distance between $\boldsymbol{T}_1^1$, $\boldsymbol{T}_1^2$, and $\boldsymbol{T}_1^3$ is constrained by the scene size, which is usually tens to hundreds of meters, while the distance between radar platforms and the scene is much greater. Setting three points $\boldsymbol{T}_1^1$, $\boldsymbol{T}_1^2$, and $\boldsymbol{T}_1^3$ along the azimuth will make $\boldsymbol{P}_n^{fake}$ and $\boldsymbol{P}_n$ locate on both sides of the scene. On the other hand, considering the precondition that distance between $\boldsymbol{P}_n'$ and $\boldsymbol{P}_n$ is in the range of positioning accuracy, about centimeter level, which is much smaller than the distance between $\boldsymbol{P}_n$ and $\boldsymbol{P}_n^{fake}$, thus the interference of $\boldsymbol{P}_n^{fake}$ can be easily avoided by setting a threshold.

Then solve for the rotation parameters θ_n^l and ϕ_n^l. According to Fig. 3, $\boldsymbol{v}_n^l$ and $\boldsymbol{P}_n^l - \boldsymbol{T}_1$ can be expressed as

$$\boldsymbol{v}_n^l = \left|\left|\boldsymbol{v}_n^l\right|\right| \left(\cos\phi_n^l \sin\theta_n^l, \cos\phi_n^l \cos\theta_n^l, \sin\phi_n^l\right) \tag{23}$$

$$\boldsymbol{P}_n^l - \boldsymbol{T}_1 = \left(x_{\boldsymbol{T}_1} - x_n^l, y_{\boldsymbol{T}_1} - y_n^l, -z_n^l\right). \tag{24}$$

Substitute (23) and (24) into (19)

$$\begin{bmatrix} x_{\boldsymbol{T}_1} - x_n^l \\ y_{\boldsymbol{T}_1} - y_n^l \\ -z_n^l \end{bmatrix}^T \begin{bmatrix} \cos\phi_n \sin\theta_n \\ \cos\phi_n \cos\theta_n \\ \sin\phi_n \end{bmatrix} = \frac{D_n \lambda \left|\left|\boldsymbol{P}_n^i - \boldsymbol{T}_1\right|\right|}{\left|\left|\boldsymbol{v}_n^l\right|\right|}. \tag{25}$$

Again, substitute three points $\boldsymbol{T}_1^1(x_{\boldsymbol{T}_1^1}, y_{\boldsymbol{T}_1^1})$, $\boldsymbol{T}_1^2(x_{\boldsymbol{T}_1^2}, y_{\boldsymbol{T}_1^2})$, and $\boldsymbol{T}_1^3(x_{\boldsymbol{T}_1^3}, y_{\boldsymbol{T}_1^3})$ into (25)

$$\begin{cases} \boldsymbol{M}_n^1 \left[\cos\phi_n \sin\theta_n, \cos\phi_n \cos\theta_n, \sin\phi_n\right]^T = D_n^1 \\ \boldsymbol{M}_n^2 \left[\cos\phi_n \sin\theta_n, \cos\phi_n \cos\theta_n, \sin\phi_n\right]^T = D_n^2 \\ \boldsymbol{M}_n^3 \left[\cos\phi_n \sin\theta_n, \cos\phi_n \cos\theta_n, \sin\phi_n\right]^T = D_n^3 \end{cases} \tag{26}$$

where $D_n^i = D_n \lambda \|\boldsymbol{P}_n^l - \boldsymbol{T}_1^i\| / \|\boldsymbol{v}_n^l\|$ and $\boldsymbol{M}_n^i = [x_{\boldsymbol{T}_1^i} - x_n^l, y_{\boldsymbol{T}_1^i} - y_n^l, z_{\boldsymbol{T}_1^i} - z_n^l]$, $i = 1, 2, 3$. Thus, the trajectory rotation amount θ_n and ϕ_n can be solved by (26). Multiple sets of points $\boldsymbol{T}_1^1$, $\boldsymbol{T}_1^2$, and $\boldsymbol{T}_1^3$ can be used to compute the trajectory rotations and offsets separately, and the final result can be obtained by averaging these multiple computations.

C. MSAR Image Fusion

In MSAR image fusion, reimage of every receiver's echo using the reconstructed trajectory is carried out in a global polar coordinate system (GPCS). The center of the first receiver trajectory is set as pole $\boldsymbol{O}'$, and the Y-axis is the polar axis. θ and r represent polar angle and polar diameter, respectively. Thus, the point target $\boldsymbol{T}$ is located at $(r_{\boldsymbol{T}}, \theta_{\boldsymbol{T}})$. Then, the nth receiver's coarse-resolution image is generated as

$$\boldsymbol{I}_n(r, \theta) = \int_{-\frac{T_a}{2}}^{\frac{T_a}{2}} \boldsymbol{S}_n^{rc}(\tau, \eta) \cdot \exp\left(j \frac{2\pi f_c R_{n,\eta}^{bi}(r, \theta)}{c}\right) d\eta \tag{27}$$

where $R_{n,\eta}^{bi}(r, \theta) = R_{1,\eta}(r, \theta) + R_{n,\eta}(r, \theta)$ denotes the range history of target (r, θ).

The angular fast Fourier transform (FFT) and range FFT are applied to each image to get the nth receiver's spatial spectrum of MSAR

$$\begin{aligned} &\boldsymbol{I}_n(K_r, K_\theta) \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \boldsymbol{I}_n(r, \theta) \exp(-jrK_r - j\theta K_\theta) \, dr d\theta \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \int_{-\frac{T_a}{2}}^{\frac{T_a}{2}} \sigma_{\boldsymbol{T}} \mathrm{rect}\left(\frac{\eta}{T_a}\right) \mathrm{sinc}\left(B\left(\tau - \frac{R_{n,\eta}^{bi}}{c}\right)\right) \\ &\quad \times \exp\left(-j \frac{2\pi f_c R_{n,\eta}^{bi}}{c}\right) \exp\left(j \frac{2\pi f_c R_{n,\eta}^{bi}(r, \theta)}{c}\right) \\ &\quad \times \exp(-jrK_r - j\theta K_\theta) \, d\eta dr d\theta \end{aligned} \tag{28}$$

where K_r and K_θ denotes the range and angular wavenumber, respectively.

According to the principle of stationary phase (POSP) [27], $\boldsymbol{I}_n(K_r, K_\theta)$ is nonzero only when

$$\frac{\partial \phi}{\partial \theta} = 0, \quad \frac{\partial \phi}{\partial r} = 0 \tag{29}$$

where ϕ is expressed as follows:

$$\phi = -\frac{2\pi f_c \left(R_{n,\eta}^{bi} - R_{n,\eta}^{bi}(r, \theta)\right)}{c} - rK_r - \theta K_\theta. \tag{30}$$

K_θ and K_r of the nth receiver is calculated as follows:

$$K_\theta^n = \frac{2\pi f_c}{c} \cdot \frac{\partial R_{n,\eta}^{bi}(r, \theta)}{\partial \theta}$$

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$$K_r^n = \frac{2\pi f_c}{c} \cdot \frac{\partial R_{n,\eta}^{bi}(r,\theta)}{\partial r}. \tag{31}$$

Also, the range history of the target can be expressed as

$$\begin{aligned} R_{n,\eta}^{bi}(r,\theta) &= R_{1,\eta}(r,\theta) + R_{n,\eta}(r,\theta) \\ &= \left\| \boldsymbol{P}_{1,\eta} \boldsymbol{T}(r,\theta) \right\| + \left\| \boldsymbol{P}_{n,\eta} \boldsymbol{T}(r,\theta) \right\| \end{aligned} \tag{32}$$

where $\boldsymbol{P}_{n,\eta} = [x_n(\eta), y_n(\eta), z_n(\eta)]$ denotes the nth receiver's position at azimuth time η.

Then, substituting (32) into (31) can get

$$\begin{aligned} K_\theta^n &= \frac{2\pi f_c}{R_T(\eta,r,\theta)\, c} \frac{\partial \boldsymbol{T}(r,\theta)}{\partial \theta} \left(\boldsymbol{T}(r,\theta) - \boldsymbol{P}_{1,\eta} \right)^T \\ &\quad + \frac{2\pi f_c}{R_R(\eta,r,\theta)\, c} \frac{\partial \boldsymbol{T}(r,\theta)}{\partial \theta} \left(\boldsymbol{T}(r,\theta) - \boldsymbol{P}_{n,\eta} \right)^T \\ K_r^n &= \frac{2\pi f_c}{R_T(\eta,r,\theta)\, c} \frac{\partial \boldsymbol{T}(r,\theta)}{\partial r} \left(\boldsymbol{T}(r,\theta) - \boldsymbol{P}_{1,\eta} \right)^T \\ &\quad + \frac{2\pi f_c}{R_R(\eta,r,\theta)\, c} \frac{\partial \boldsymbol{T}(r,\theta)}{\partial r} \left(\boldsymbol{T}(r,\theta) - \boldsymbol{P}_{n,\eta} \right)^T \end{aligned} \tag{33}$$

where target $\boldsymbol{T}(r,\theta)$ in the polar coordinate system denotes

$$\boldsymbol{T}(r,\theta) = \begin{bmatrix} T_x(r,\theta) \\ T_y(r,\theta) \\ T_z(r,\theta) \end{bmatrix}^T = \begin{bmatrix} x_n(\eta) + r\sin\alpha\sin\theta \\ y_n(\eta) + r\cos\theta \\ z_n(\eta) - r\cos\alpha\sin\theta \end{bmatrix}^T \tag{34}$$

where α is the angle between vector $\boldsymbol{OT}$ and the YOZ plane. Also, thus $\partial \boldsymbol{T}(r,\theta)/\partial\theta$, $\partial \boldsymbol{T}(r,\theta)/\partial r$ in (33) can be calculated as

$$\begin{aligned} \frac{\partial \boldsymbol{T}(r,\theta)}{\partial \theta} &= \left[r\sin\alpha\cos\theta, -r\sin\theta, -r\cos\alpha\cos\theta \right] \\ \frac{\partial \boldsymbol{T}(r,\theta)}{\partial r} &= \left[\sin\alpha\sin\theta, \cos\theta, -\cos\alpha\sin\theta \right]. \end{aligned} \tag{35}$$

According to the expression of K_θ^n, K_r^n, the range of the subaperture spatial spectrum of each coarse resolution image is calculated

$$\begin{aligned} K_\theta^n &\in \left[\min_{\theta,r,\eta} K_\theta^n, \max_{\theta,r,\eta} K_\theta^n \right] \\ K_r^n &\in \left[\min_{\theta,r,\eta} K_r^n, \max_{\theta,r,\eta} K_r^n \right]. \end{aligned} \tag{36}$$

Also, the spatial spectrum center of the nth receiver denotes

$$\begin{aligned} K_\theta^{n,c} &= \frac{2\pi f_c}{c} \cdot \left. \frac{\partial R_{n,\eta_c}^{bi}(r_T,\theta)}{\partial \theta} \right|_{r=r_T} \\ K_r^{n,c} &= \frac{2\pi f_c}{c} \cdot \left. \frac{\partial R_{n,\eta_c}^{bi}(r,\theta_T)}{\partial r} \right|_{\theta=\theta_T}. \end{aligned} \tag{37}$$

For the sake of computational efficiency, the range of the angular spatial spectrum of the each receiver's image, ΔK_θ, is slightly larger than the range of its spatial spectrum, $\Delta K_\theta^n = \max_{\theta,r,\eta} K_\theta^n - \min_{\theta,r,\eta} K_\theta^n$. So the spatial spectrum of the subaperture images is periodically extended and maybe folded. It is necessary to first judge whether the spatial spectrum is folded according to the spatial spectrum distribution. The spatial spectrum region of the nth receiver denotes $[\min_{\theta,r,\eta} K_\theta^n, \max_{\theta,r,\eta} K_\theta^n]$ and the angular spatial spectrum region of a receiver's image is $[-\Delta K_\theta/2, \Delta K_\theta/2]$. Spectral folding occurs when there exists an arbitrary integer N_k such that both $\min_{\theta,r,\eta} K_\theta^n < (2N_k+1)\Delta K_\theta/2$ and $\max_{\theta,r,\eta} K_\theta^n > (2N_k+1)\Delta K_\theta/2$ hold simultaneously. In this case, shift the nth receiver's echo spatial spectrum center to 0 to achieve unfolding. Then, shift the spatial spectrum to its real position according to (37), that is, shift the nth receiver's echo spatial spectrum center from 0 to $K_\theta^{n,c}$. Therefore, unfold and shift all subaperture images to align the spatial spectra of all receivers' images to their respective true positions, then all receivers' images are fused in the spatial spectrum domain. Performing 2-D IFFT can get the MSAR full-resolution imaging results.

D. Computational Complexity Analysis

To present the efficiency clearly, the computational complexity of the proposed method is analyzed. Let the SAR image size be $N_i \times N_i$, the azimuth sampling number be N_A, and the receiver number be N. The overall workflow consists of three main stages: Coarse image Formation, MSAR trajectory reconstruction, and MSAR image fusion.

Coarse image formation performs BP of each receiver's echoes with an individual complexity of $(N_i \times N_i \times N_A)/N^2$, resulting in a total computational complexity of $O_{S1} = (N_i \times N_i \times N_A)/N$.

The MSAR trajectory reconstruction process comprises multiple steps. First, the autofocus stage requires iteratively repeating the imaging process; its complexity is dominated by the kth iterative imaging operation itself, denoted as $k(N_i \times N_i \times N_A)/N^2$. Subsequently, the complexity of the registration step is dominated by the image size, scaling as $(N_i \times N_i)/N$, which is significantly lower than that of imaging. The computational cost of error parameter inversion primarily involves equation solving; given the orders-of-magnitude difference compared with imaging, this cost is considered negligible. Finally, the complexity of reimaging using the reconstructed trajectory for each receiver is $(N_i \times N_i \times N_A)/N^2$. Thus, $O_{S2} = k(N_i \times N_i \times N_A)/N + (N_i \times N_i \times N_A)/N$.

Regarding MSAR image fusion, the process entails performing a Fourier transform on each subimage, establishing the fusion coordinate system, and applying an inverse Fourier transform to the fused spatial spectrum. The computational complexities of these three steps are $N_i^2/N\log(N_i^2/N)$, N_i^2, and $N_i^2\log(N_i^2)$, respectively. The complexity of MSAR image fusion is $O_{S3} = N_i^2\log(N_i^2/N) + N_i^2 + N_i^2\log(N_i^2)$.

In total, the proposed method needs a computational complexity of $O_{S1} + O_{S2} + O_{S3} = ((k+2)N_i^2 N_A)/N + N_i^2(4logN_i - logN + 1)$. It can be observed that as the number of platforms N increases, the computational load of the proposed algorithm decreases significantly.

IV. Numerical Simulation and Experiment

In this section, the effectiveness of the proposed method is verified with numerical simulations and field-measured MSAR data, which is from the three UAV-based MSAR sequential imaging experiment conducted in 2025.

A. High-Precision MSAR Sequential Imaging Simulation

In numerical simulations, a ground area with several strong point targets is adopted as the simulation scene to show the algorithm's effectiveness, as shown in Fig. 7. The scene size is 320 × 320 m. To evaluate imaging quality, three isolated

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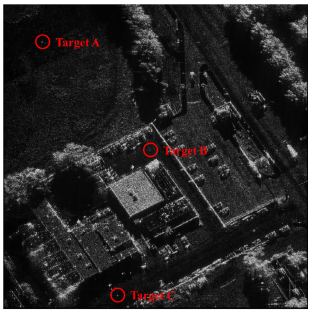


Fig. 7. Original simulation scene.

TABLE II
SIMULATION PARAMETERS

Parameters	Value
Carrier frequency	9.5 GHz
Bandwidth	1 GHz
Pulse repetition frequency	1200 Hz
Resolution	0.3 m
Center range	5000 m
Platform velocity	86 km/h
Platform number	16

strong point targets were introduced into the scene, denoted as A–C, located at $[-40, 148\text{m}]$, $[-6, -6\text{m}]$, $[-120, -120\text{m}]$, respectively.

Detailed simulation parameters are given in Table II.

For contrast, we adopt several methods to image the MSAR echo.

1) *TDA Method:* Use time-domain imaging method [13] to image the target according to the ideal trajectory; the imaging result reflects the influence of trajectory error on the image.
2) *BP+MSA:* MSA proposed by Ash [25] is adopted to implement pulse-wise phase correction based on maximizing the image sharpness criterion after BP processing. The number of MSA iterations is set to 3.
3) *BP+MEE:* The motion error estimation algorithm (MEE) proposed in [23], which first reconstructs each receiver's actual trajectory and then performs autofocus to compensate for high-order motion error is adopted. Since MEE is proposed for monostatic SAR, we first calculate the equivalent monostatic configuration [28] formed by the first receiver and each receiver in MSAR and then perform MEE for the equivalent configuration. Also, the number of autofocus iterations is set to 3.
4) *BP+GEC:* A multistatic geometry error compensation method proposed in [14] is adopted to estimate trajectory error. The differential evolution optimization within GEC is configured with a population size of 500, a scaling factor of 0.5, a crossover probability of 0.1, and a maximum of 500 generations.
5) *The Proposed Method:* The proposed method is carried out to reconstruct the actual trajectories of the multiple platforms in MSAR, and a high-resolution, high-precision image is formed by coherently fusing multiple receivers' echoes.

In numerical simulations, imaging is performed separately for all receivers' echoes. Fig. 8(a)–(c) shows the first, second, and third receivers' images, with targets A–C magnified, exhibiting significant defocusing and noticeable spatial offsets. After trajectory error compensation, the results from all receivers are well-focused and properly aligned, as shown in Fig. 8(d)–(f). It is shown that the image of an individual receiver in MSAR does not have sufficient azimuth resolution due to the short aperture.

MSAR high-precision trajectory reconstruction results of the proposed method are given and analyzed in Figs. 9 and 10. The nominal trajectory, the preset trajectory with error, and the reconstructed trajectory are plotted in Fig. 9. It can be seen that the reconstructed trajectories coincide very well with the preset trajectories.

In addition, Fig. 10 gives MSAR trajectory phase error before and after trajectory reconstruction. The 16 different receivers' phase errors are depicted with different colors and lines. It can be seen that under the preset trajectory, the phase error of MSAR sequential imaging is significant and obviously crosses a range cell. While the phase error introduced by the trajectory reconstructed is less than $\pi/4$, the error will not cross the range cell, ensuring high-accuracy imaging ability of MSAR.

The imaging results, magnification of preset strong points, are presented in Fig. 11. The image result of the TDA algorithm is shown in Fig. 11(a), which is seriously defocused because the multiplatform trajectory error affects the coherent fusion of all receivers' images. MSA's image result has a better focusing effect, as plotted in Fig. 11(b). However, its per-pulse phase error compensation strategy has inherent limitations: it cannot handle envelope errors and struggles to effectively compensate for spatial-variant trajectory errors under large observation angles. Consequently, its overall imaging quality remains unsatisfactory. The MEE method attempts to reconstruct the trajectory of each receiver in MSAR; however, since it estimates trajectory segments based solely on slope and intercept, it fails to accurately recover the relative distribution relationship among the receivers. This limitation leads to the blurred and defocused imaging result shown in Fig. 11(c). Although the GEC method introduces an error model capable of fully describing MSAR errors, it relies on a search strategy to estimate all error parameters. As a result, its estimation accuracy is highly sensitive to the initial values and search intervals. Furthermore, this optimization algorithm is prone to falling into local optima, causing parameter estimation deviations and poor imaging results, as depicted in Fig. 11(d). In contrast, the proposed method demonstrates superior performance. As shown in Fig. 9, its trajectory estimation results align closely with the preset trajectories, achieving high-precision estimation of multiplatform trajectory errors. Benefiting from this accurate estimation, the final imaging result of the proposed method, presented in Fig. 11(e), exhibits excellent focusing performance. This validates the significant advantages and robustness of the proposed algorithm in MSAR sequential imaging processing.

To demonstrate the improvement in imaging quality, particularly in resolution enhancement, achieved by increasing

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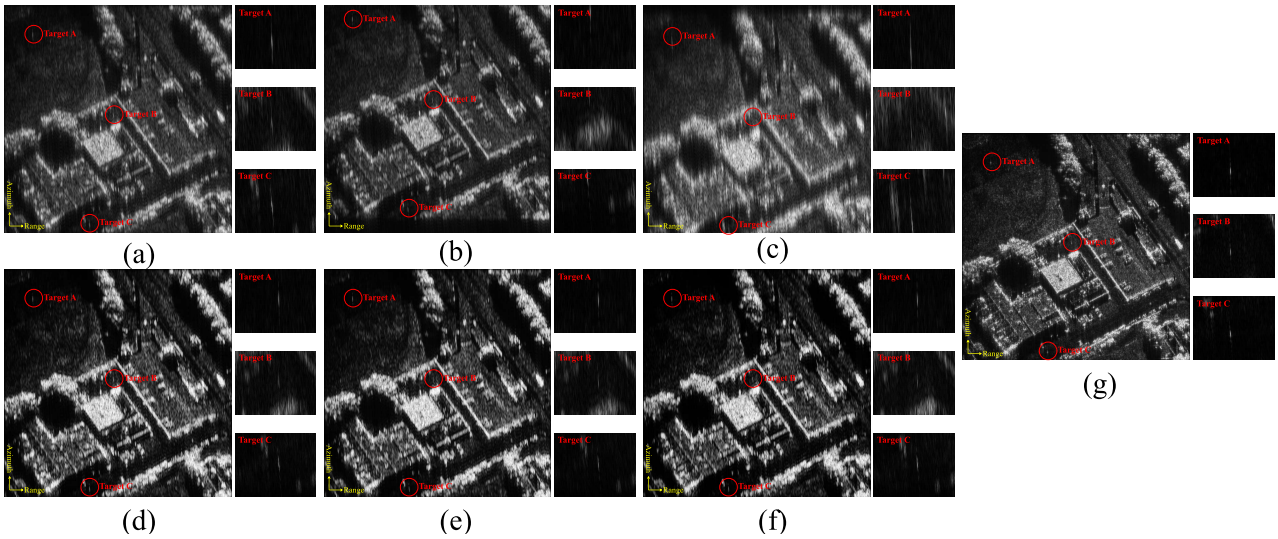


Fig. 8. Image results of different receivers before and after trajectory error compensation. (a) First receiver. (b) Second receiver. (c) Third receiver. (d) First receiver after trajectory error compensation. (e) Second receiver after trajectory error compensation. (f) Third receiver after trajectory error compensation. (g) Fused image result of first–third receivers using the proposed method.

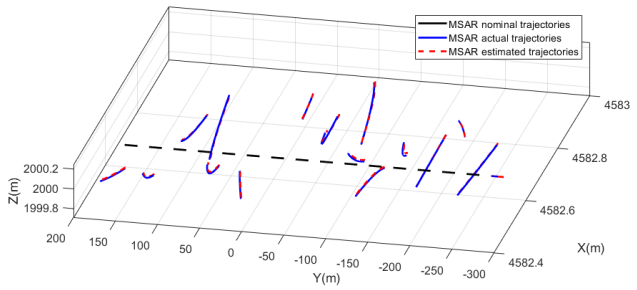


Fig. 9. MSAR trajectory reconstruction result.

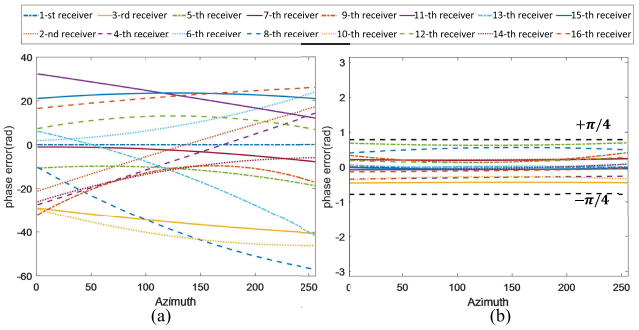


Fig. 10. MSAR phase error (a) before and (b) after trajectory reconstruction.

the number of platforms in MSAR sequential imaging, fused imaging of receivers 1–3 is performed, yielding the imaging results shown in Fig. 8(g). It can be observed that after processing with the proposed method, the data from the three receivers are effectively coherently fused. Within the same flight duration, compared with Fig. 8(d)–(f), the resolution of the fused imaging from three receivers is significantly higher than that of a single receiver, yet compared with Fig. 11(e), still considerably lower than that achieved with sixteen receivers. This clearly demonstrates that a greater number of platforms leads to higher resolution, proving the critical importance of platform number for the imaging capability of sequential SAR systems.

For quantitatively comparing the performance of the above imaging methods, quality indexes [25], [29], [30], including image sharpness Q_S, entropy Q_E, and contrast Q_C, are adopted and listed in Table III, which are defined as

$$Q_S = \sum_i \Psi(\boldsymbol{Im}(i))^2$$

$$Q_E = -1/E_z \sum_i \|\boldsymbol{Im}(i)\|^2 \ln \|\boldsymbol{Im}(i)\|^2 + \ln E_z$$

$$Q_C = \sqrt{A[\Psi(\boldsymbol{Im}(i)) - A(\Psi(\boldsymbol{Im}(i)))]}/A(\Psi(\boldsymbol{Im}(i))) \quad (38)$$

where $\boldsymbol{Im}(i)$ is the value of ith image pixel, $\Psi(\boldsymbol{Im}(i)) = \boldsymbol{Im}(i)\boldsymbol{Im}(i)^*$ is the intensity of ith pixel, $E_z = \sum_i \|\boldsymbol{Im}(i)\|^2$, and $A(\cdot)$ is the arithmetic mean of the samples. Also, the 3-dB impulse response width (IRW), peak sidelobe ratio (PSLR), and integrated sidelobe ratio (ISLR) of selected scattering point targets labeled in Fig. 11 are also listed. Profiles of the scattering point targets are compared in Fig. 12. Although the PSLR and ISLR are slightly inferior to the theoretical values, the best quantitative quality index values could be achieved using the proposed method when performing the MSAR sequential imaging.

The computational times of the various algorithms on a 13th Gen Intel[1] Core[2] i7-13700KF (3.4 GHz) platform are also evaluated.The computational times of TDA, BP+MSA, BP+MEE, BP+GEC and the proposed method are 55, 3201, 4002, 2845, and 255 s, respectively. TDA performs image formation only, without any compensation processing, and thus achieves the lowest computational time. In contrast, BP+MSA, BP+MEE, and BP+GEC require significantly higher computational costs due to iterative optimization and repeated imaging. The proposed method demonstrates moderate computational

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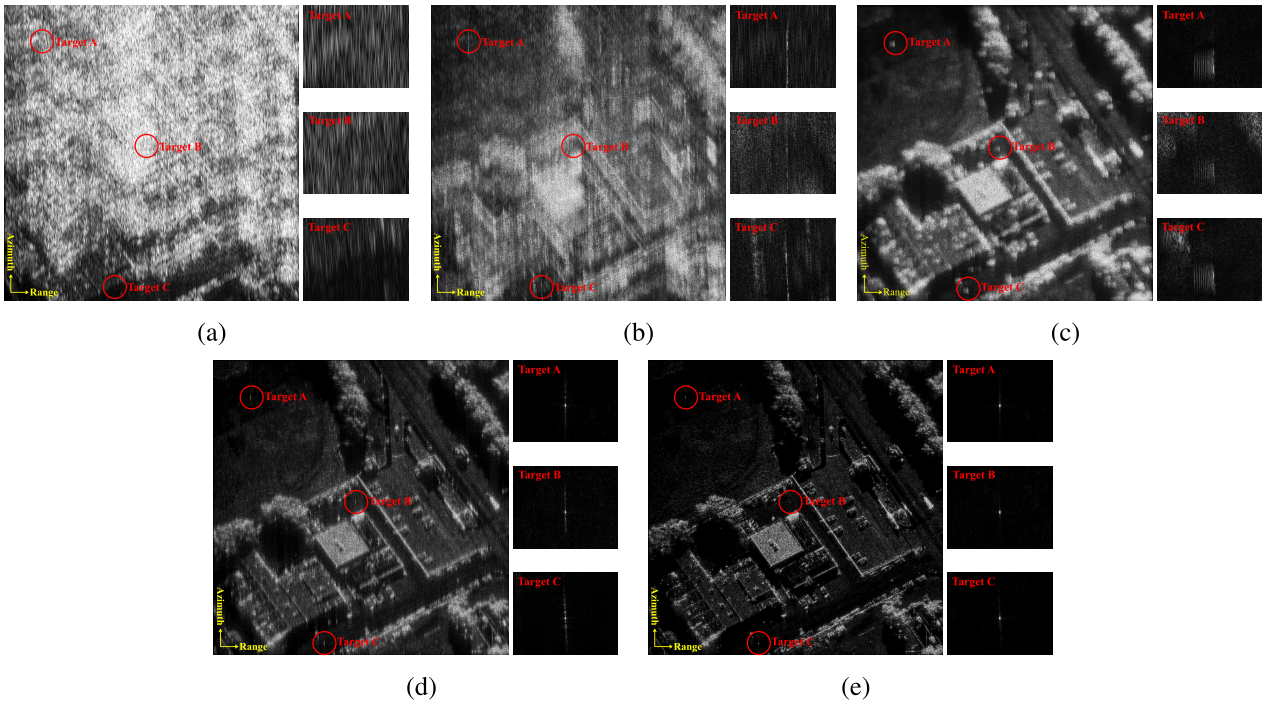


Fig. 11. Total 16-platform image fusion results of different methods. (a) TDA [13]. (b) BP+MSA [25]. (c) BP+MEE [23]. (d) BP+GEC [14]. (e) Proposed method.

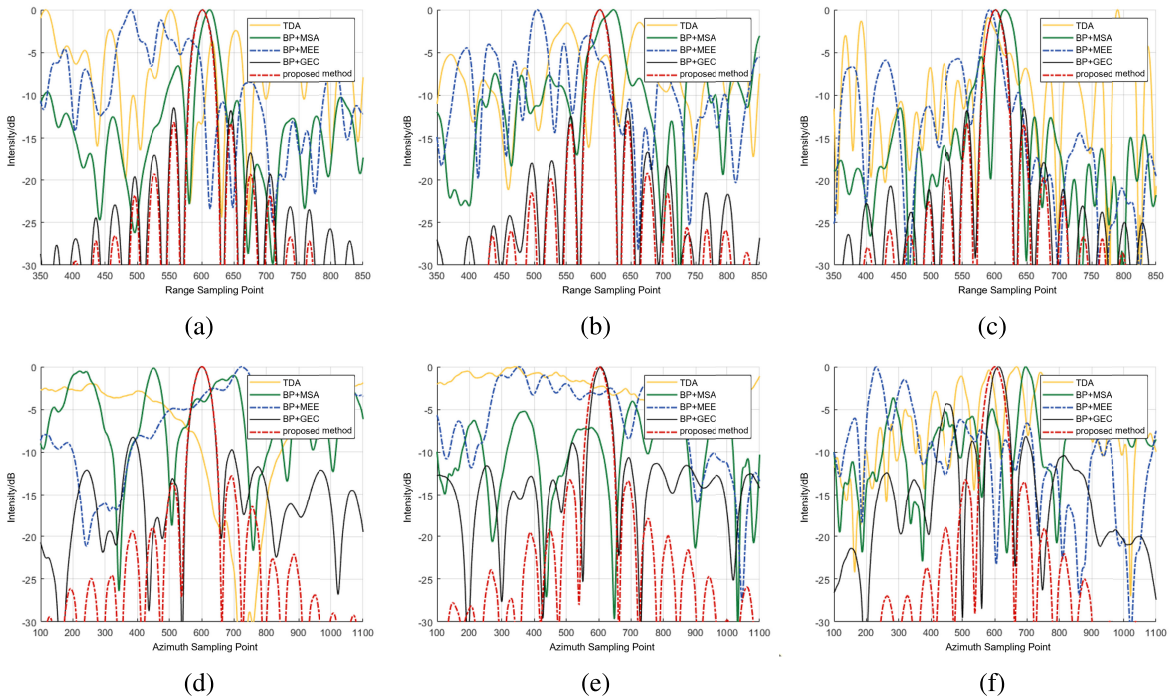


Fig. 12. Comparison of profiles of image results of different methods, including range profile of (a) Target A, (b) Target B, and (c) Target C, and azimuth profile of (d) Target A, (e) Target B, and (f) Target C.

complexity while delivering the best imaging performance among all compared approaches.

B. High-Precision MSAR Sequential Imaging Experiment

To validate the effectiveness of the subaperture spatial spectrum fusion method, an MSAR imaging experiment is conducted. The optical image of the experimental scenario and experimental system is depicted in Fig. 13.

The experiment adopts three UAVs on which a compact radar system is mounted. Each radar system comprises not only the radar antenna but also additional equipment, including GPS, RTK antenna, INS module, and synchronization module. The INS module provides positioning data, while the

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TABLE III

QUANTITATIVE ANALYSIS OF THE SIMULATION DATA IMAGING RESULT (SP: STRONG POINT)

Imaging algorithm	SP	Range profile IRW (m)	Range profile PSLR (dB)	Range profile ISLR (dB)	Azimuth profile IRW (m)	Azimuth profile PSLR (dB)	Azimuth profile ISLR (dB)	Full image Q_S	Full image Q_E	Full image Q_C
Group 1: Individual Receiver Imaging										
Receiver 1	A	0.17	-13.45	-10.06	5.48	-11.10	-7.47	0.93	16.21	1.00
	B	0.16	-13.45	-10.34	5.36	-12.35	-5.37			
	C	0.17	-12.75	-10.47	5.52	-17.62	-8.95			
Receiver 2	A	0.17	-13.20	-10.58	5.57	-10.50	-6.76	0.97	16.16	0.99
	B	0.16	-13.58	-9.99	5.52	-8.92	-5.79			
	C	0.17	-13.68	-10.18	5.62	-13.60	-6.35			
Receiver 3	A	0.17	-13.07	-10.15	7.62	-9.80	-3.32	0.68	16.38	0.98
	B	0.17	-13.29	-10.32	7.31	-9.80	-2.96			
	C	0.18	-12.92	-10.28	7.67	-10.08	-4.20			
Group 2: Individual Receiver Imaging after trajectory error compensation										
Receiver 1	A	0.17	-13.64	-10.12	4.57	-13.57	-10.76	1.02	16.14	1.01
	B	0.16	-13.34	-10.08	4.53	-11.80	-11.07			
	C	0.16	-13.65	-10.53	4.47	-14.35	-10.54			
Receiver 2	A	0.16	-12.55	-10.06	4.49	-11.30	-10.22	0.98	16.15	1.00
	B	0.16	-13.76	-10.92	4.54	-12.12	-10.16			
	C	0.17	-13.00	-10.19	4.57	-11.25	-12.71			
Receiver 3	A	0.17	-13.77	-10.18	4.53	-12.13	-10.88	1.00	16.15	1.00
	B	0.18	-12.94	-10.38	4.47	-10.21	-9.95			
	C	0.17	-13.46	-10.67	4.50	-10.76	-10.24			
Group 3: Fused Imaging (Receivers 1-3)										
Proposed Method	A	0.18	-13.29	-10.21	1.45	-13.32	-10.07	14.77	15.85	1.07
	B	0.17	-13.35	-10.39	1.36	-13.59	-10.04			
	C	0.18	-13.21	-10.04	1.43	-13.40	-10.85			
Group 4: Total 16-Receiver Imaging Comparison										
TDA [13]	A	—— defocused ——			—— defocused ——			51.05	16.84	1.09
	B	—— defocused ——			—— defocused ——					
	C	—— defocused ——			—— defocused ——					
BP+MSA [25]	A	—— defocused ——			—— defocused ——			1000.10	16.79	1.15
	B	—— defocused ——			—— defocused ——					
	C	—— defocused ——			—— defocused ——					
BP+MEE [23]	A	—— defocused ——			—— defocused ——			105.67	16.70	1.15
	B	—— defocused ——			—— defocused ——					
	C	—— defocused ——			—— defocused ——					
BP+GEC [14]	A	0.20	-11.43	-10.18	0.28	-9.69	-2.66	475.28	15.95	1.16
	B	0.19	-11.31	-9.85	0.28	-8.88	-1.47			
	C	0.20	-11.58	-9.96	0.29	-8.14	-0.31			
Proposed Method	A	0.17	-13.29	-10.21	**0.28**	-13.22	-10.07	**1075.58**	**15.43**	**1.18**
	B	0.17	-13.35	-10.39	**0.27**	-13.19	-10.04			
	C	0.17	-13.21	-10.04	**0.27**	-13.70	-10.85			

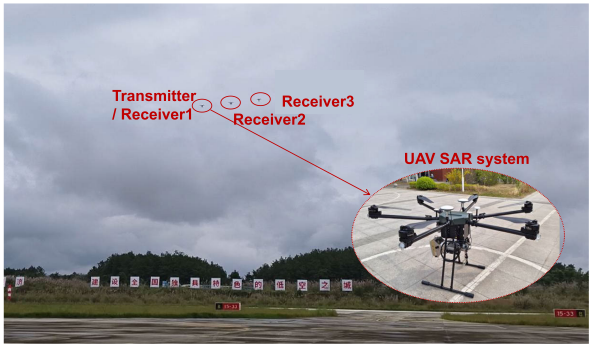


Fig. 13. Experimental scene.

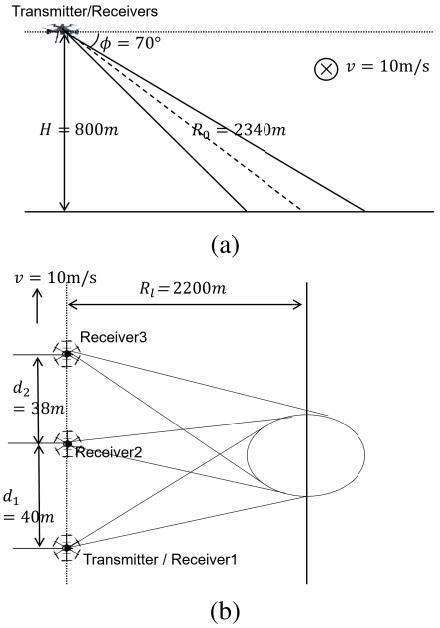


Fig. 14. Experiment configuration. (a) Side view. (b) Top view.

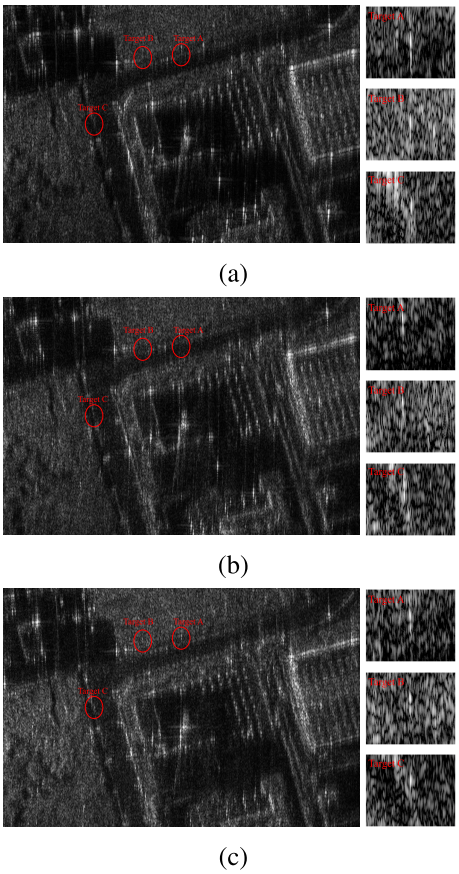
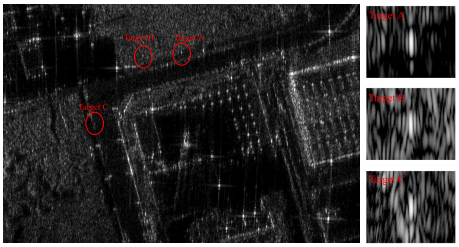


Fig. 15. Image results of all receivers. (a) First receiver. (b) Second receiver. (c) Third receiver.

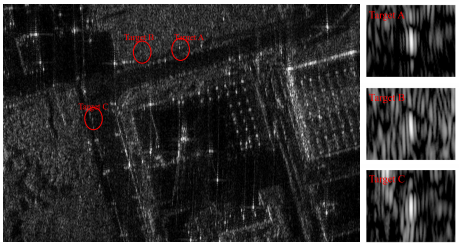
synchronization module delivers high-precision synchronization pulses. Triggered by these high-precision pulses, the transmitter and receivers operate at a fixed synchronized PRF.

The specific experiment geometry is shown in Fig. 14. In the experiment, three UAV-borne SAR serve as transmitter and receiver, respectively. They fly in the same direction, one ahead and one behind along the heading, with the transmitter operating in self-transmit–self-receive mode and the receivers solely in receive mode, both functioning in spotlight mode. The

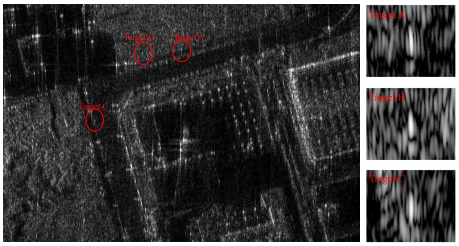
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(a)



(b)



(c)

Fig. 16. Image results of all receivers after trajectory error compensation. (a) First receiver. (b) Second receiver. (c) Third receiver.

carrier frequency and bandwidth of the system are 9.5 GHz and 800 MHz, respectively. Also, the system PRF is 2000 Hz. The transmitter and receivers fly with an along-track separation of approximately 40 m and at altitudes of 780, 800, and 820 m. The flight speed is 10 m/s and the operation distance is 2340 m. The synthetic aperture time is 1.5 s, thus the 3-UAV MSAR system's theoretical azimuth resolution is 0.55 m. Effectively fusing all receivers' echoes to achieve resolution enhancement verifies the efficacy of the method proposed in this article.

After acquiring the MSAR echo, the direct wave is first extracted from the receiver echo, and direct wave compensation is performed to eliminate the negative impact of synchronization error on imaging [31], [32]. Subsequently, imaging is performed separately for all receivers' echoes. As shown in Fig. 15, all receivers' imaging results exhibit noticeable defocusing and misalignments. After trajectory error compensation, the results from all receivers achieve well-focused and aligned results, as shown in Fig. 16.

Next, registration is performed, and trajectory reconstruction is conducted based on the registration results. The transmitter and receiver echoes were then reimaged and rapidly fused in the spatial spectral domain, yielding the imaging results of the proposed method as shown in Fig. 17(e). Concurrently, comparative experiments, including TDA, BP+MSA, BP+MEE, and BP+GEC, are conducted to process the MSAR echoes,

TABLE IV

QUANTITATIVE ANALYSIS OF THE EXPERIMENTAL DATA IMAGING RESULT (SP: STRONG POINT)

Imaging algorithm	SP	Range profile			Azimuth profile			Full image		
		IRW (m)	PSLR (dB)	ISLR (dB)	IRW (m)	PSLR (dB)	ISLR (dB)	Q_S	Q_E	Q_C
Group 1: Individual Receiver Imaging										
Receiver1	A	0.28	-10.46	-6.32	2.58	-5.37	-2.70	4.05	13.52	1.02
	B	0.26	-6.62	-1.82	2.98	-5.38	0.10			
	C	0.38	-4.82	-3.87	1.95	-4.41	1.98			
Receiver2	A	0.26	-7.80	-2.84	1.79	-3.93	-1.77	3.31	13.74	1.02
	B	0.26	-8.72	-2.66	1.78	-5.38	-0.04			
	C	0.24	-5.16	-2.27	3.33	-9.69	-0.08			
Receiver3	A	0.26	-9.15	-3.18	3.52	-2.81	-1.00	0.42	13.96	0.98
	B	0.30	-12.25	-5.61	1.79	-3.33	0.05			
	C	0.28	-7.73	-3.16	2.28	-10.30	-5.13			
Group 2: Individual Receiver Imaging after trajectory error compensation										
Receiver1	A	0.28	-10.66	-7.02	1.78	-10.28	-5.57	7.98	13.24	1.04
	B	0.26	-7.26	-2.62	1.77	-12.20	-3.58			
	C	0.27	-7.62	-4.69	1.79	-6.56	-4.40			
Receiver2	A	0.26	-8.93	-4.84	1.71	-7.44	-6.86	3.54	13.61	1.02
	B	0.25	-10.41	-4.96	1.72	-9.47	-4.66			
	C	0.23	-7.35	-3.85	1.78	-11.09	-5.97			
Receiver3	A	0.26	-10.51	-4.08	1.82	-8.96	-4.51	1.00	13.91	1.00
	B	0.30	-13.47	-6.25	1.76	-6.45	-4.47			
	C	0.27	-9.35	-4.93	1.75	-14.78	-9.27			
Group 3: Total 3-Receiver Imaging Comparison										
TDA [13]	A	0.28	-7.61	-3.07	1.49	-2.81	-0.44	16.61	13.79	1.05
	B	0.27	-6.62	-0.63	– defocused –					
	C	0.27	-1.63	3.30	4.35	-2.90	-5.90			
BP+MSA [25]	A	0.28	-10.53	-6.15	1.60	-7.75	-2.78	16.58	13.74	1.05
	B	0.27	-14.40	-1.18	1.33	-7.55	-0.07			
	C	0.25	-3.18	3.00	0.55	-3.41	2.73			
BP+MEE [23]	A	0.27	-11.20	-4.92	1.79	-6.57	-3.64	14.47	13.84	1.05
	B	0.27	-7.24	0.08	1.28	-6.45	-1.67			
	C	0.29	-5.20	3.95	1.53	-6.84	0.10			
BP+GEC [14]	A	0.25	-11.67	-7.75	0.42	-0.65	3.84	41.92	13.56	1.06
	B	0.26	-10.57	-4.23	0.46	-1.66	2.31			
	C	0.26	-5.15	-1.45	0.48	-1.97	3.05			
Proposed method	A	0.26	-8.44	-7.01	0.61	-8.61	-6.27	**122.48**	**13.41**	**1.08**
	B	0.27	-12.06	-8.01	0.52	-8.97	-6.90			
	C	0.26	-8.41	-6.74	0.52	-10.64	-7.48			

with the imaging results also depicted in Fig. 17. Fig. 17(a) demonstrates that the BP method yields the most severe defocusing imaging results. Following application of the MSA, MEE, and GEC methods, Fig. 17(b)–(d) exhibits significantly improved image results, with Fig. 17(e) achieving the optimal focusing effect. However, large-area imaging results provide only a rough, general description of imaging quality.

The clearly marked targets A–C are enlarged and displayed in Fig. 17, too. We also conducted quantitative analysis, calculating multiple quantitative metrics for the imaging results of each method. These included image sharpness, entropy, and contrast of the imaging areas, alongside resolution, PSLR, and ISLR metrics for the strong target A–C, all summarized in Table IV.

It is readily apparent that in the imaging results from TDA and BP+MEE, the azimuth defocusing of point targets is quite pronounced. These methods demonstrate no capacity to compensate for MSAR trajectory errors nor to enhance resolution. BP+MSA achieves a degree of focusing, yet its imaging quality shows no discernible improvement over each platform's imaging results and fails to elevate azimuth

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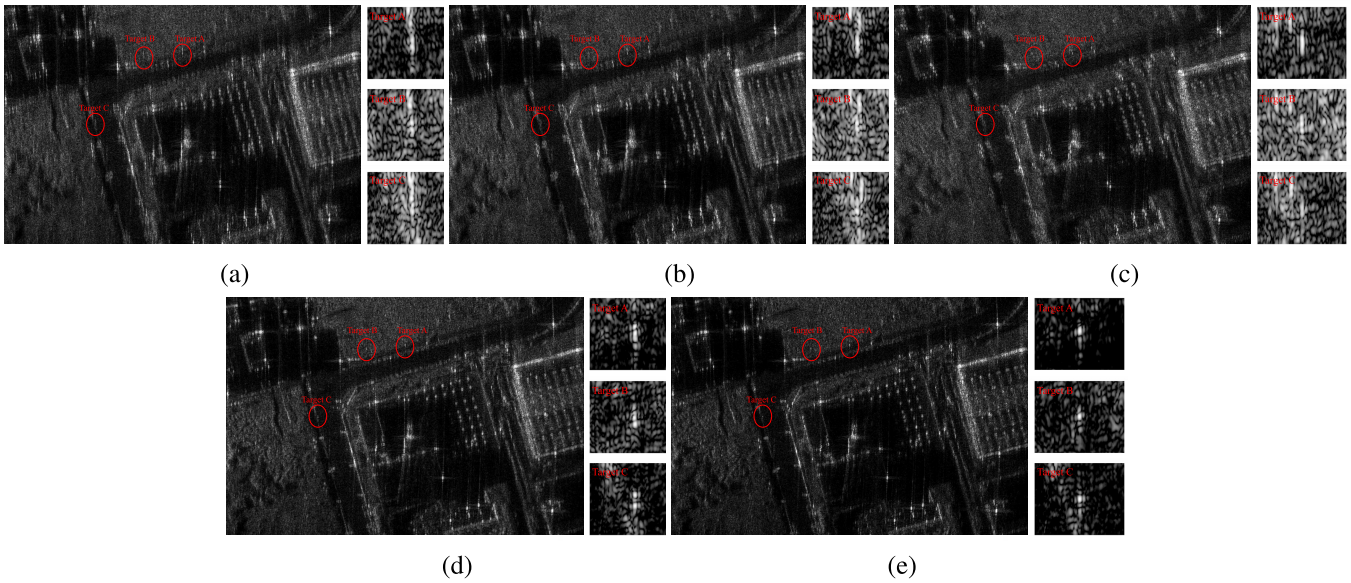


Fig. 17. Image results of different methods. (a) TDA [13]. (b) BP+MSA [25]. (c) BP+MEE [23]. (d) BP+GEC [14]. (e) Proposed method.

resolution. This indicates that the method cannot achieve coherent fusion of the MSAR spatial spectrum. BP+GEC achieves passable imaging resolution, but its imaging results suffer from obvious sidelobe lifting. This indicates the method's performance is constrained by parameter estimation accuracy. It is evident that the proposed method achieves a resolution enhancement for targets A–C, from 1.71, 1.72, and 1.75 to 0.61, 0.52, and 0.52 m, respectively, consistent with theoretical calculations.

For computational burden analysis, the computational times of TDA, BP+MSA, BP+MEE, BP+GEC, and the proposed method are 511, 6002, 7500, 5870, and 2418 s, respectively. The proposed method achieves the best imaging performance at the cost of moderate computational complexity.

The experiments in this article successfully validate the effectiveness of the proposed imaging method. Only this approach achieves an effective enhancement in image resolution that aligns with the theoretical resolution. This demonstrates the effectiveness of the proposed method.

V. Conclusion

This article investigates a one-transmitting multiple-receiving MSAR sequential imaging method under multiplatform trajectory errors. First, MSAR's echo model under multiplatform multiorder trajectory errors is constructed and analyzed, laying the foundation for MSAR coherent imaging. Also, then, a high-precision multiplatform trajectory reconstruction method is developed, enabling accurate reconstruction of flexible multiplatform trajectory errors. Finally, an MSAR sequential imaging method using subaperture spatial spectrum fusion is proposed, which compensates for MSAR trajectory errors and ensures multiplatform image coherent fusion accuracy through fast spatial spectrum domain image fusion. Furthermore, numerical simulations and processing of real measurements from a three-UAV MSAR system validate the proposed method's effectiveness in MSAR coherent data fusion and resolution enhancement.

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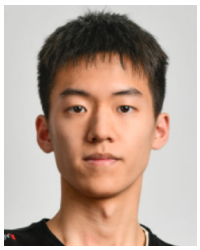
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